
Software Requirements Specification

for

Nupjuk Guide

Version 1.0 approved

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KAIST

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1. Introduction

1.1 Purpose

This document presents the software requirements for a campus guide application designed for KAIST. The system aims to support users who are unfamiliar with the campus, such as freshmen, exchange students, and visitors, by providing practical campus information in an accessible form. Existing campus information is often fragmented across multiple sources, and the current campus map does not always provide enough practical detail for users to navigate the campus effectively. The document defines the major functions of the system as well as the quality requirements that should be considered during development.

1.2 Document Conventions

1.2.1 Defined Terms

In this document, the term **user** refers to any person who uses the application to access campus information. A **manager** is a user with permission to create, modify, and remove information displayed in the system. An **administrator** is responsible for account approval and related administrative tasks.

A **marker** refers to a location-based item shown on the map. Some markers may contain **child markers**, which represent detailed spaces or facilities inside a larger location. In such cases, the larger marker is called a **parent marker**, meaning the marker to which the child markers belong. Information in the system is divided into **static information**, such as building or facility locations, and **dynamic information**, such as event notices or bus updates.

More detailed definitions of important terms are provided in **Appendix A. Glossary**.

1.2.2 Formatting Conventions

This document is written in **Times New Roman** font. Important terms and key concepts are highlighted in **bold** for clarity. Section headings and numbering are used consistently to improve readability and organization throughout the document.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following readers:

- **Developers:** to understand the required system functions, user interactions, and implementation scope.
- **Designers:** to understand the structure of the service and the expected user interface flow.
- **Testers:** to identify the expected behavior of the system and verify whether the requirements are satisfied.
- **Project Managers:** to review the project scope, major features, and overall direction of development.

For an overall understanding of the project, readers are recommended to begin with the **Introduction** and **Overall Description** sections. Readers who need detailed system behavior should refer to the **System Features** and **External Interface Requirements** sections.

1.4 Product Scope

The proposed system is a campus guide application that provides integrated, location-based information for users. The system is intended to help freshmen, exchange students, visitors, and other users who may have difficulty finding practical campus information scattered across multiple sources.

The main scope of the system includes the following:

- **Campus Map Information**
The system provides map-based information about major campus locations, including buildings, facilities, and other important places.
- **Nested Location Information**
The system supports detailed location information inside larger places, such as lecture rooms inside academic buildings or specific spaces inside student facilities.
- **Dynamic Information**
The system provides time-sensitive dynamic information including dining, bus, and temporary campus events.
- **Search and Filter Functions**
The system allows users to search for information and filter results in order to find relevant locations and services more efficiently.
- **Manager-Based Information Update**
The system allows authorized managers to create, update, and delete information displayed in the application, especially for dynamic content.

1.5 References

The following external references were consulted for certain interface and platform decisions described in this document.

1. Kakao Map URL Scheme Guide
2. Apple Developer Support documentation for iOS version compatibility
3. Amazon Web Services documentation

General reference materials on REST APIs, JSON, and HTTPS-based communication

2. Overall Description

2.1 Product Perspective

The proposed system is a campus guide application that integrates campus information into a single map-based service. It is designed to help users who are unfamiliar with the campus, such as freshmen, exchange students, and visitors, access practical information more easily. The system displays information through **markers** on a campus map. Some markers may include **child markers** to represent more specific spaces or facilities within a given location.

The system is mainly intended for general users **without requiring login**. Users can browse the map, search for locations, and view campus-related information directly. A separate web-based manager interface is provided for **authorized managers** to log in and update events or other dynamic information. Some information in the system may also depend on periodically updated external sources.

2.2 Product Functions

2.2.1 Map and Marker Functions

The system provides an interactive campus map through which users can access location-based information. Users can browse markers displayed on the map, select a marker to view detailed information, and access child markers for more specific spaces or facilities.

There are various types of campus information displayed through the map interface. This includes building and facility information, dining information, bus information, and temporary event information. Some dynamic information, such as dining or bus, may be refreshed periodically.

2.2.2 Search and Filter Functions

The system allows users to search for campus locations by keyword. It also provides filtering functions so that users can narrow results by category and find relevant places more efficiently.

2.2.3 Marker Management Functions

The system provides a separate web-based interface for authorized managers. Through this interface, managers can log in and create, update, or delete marker information displayed in the system.

2.3 User Classes and Characteristics

The system includes the following user classes:

- **General Users**

General users include freshmen, exchange students, visitors and other people who need campus information. They are expected to use the system without login. Since many users may be unfamiliar with the campus, the system should provide intuitive navigation and clear information presentation.

- **Managers**

Managers are authorized users who log in through the web interface to manage marker details. They are responsible for updating information shown in the system.

Apart from user classes above, the system also includes a separate **administrator**. The administrator is responsible for overall system administration, including reviewing and approving manager account requests, handling password reset requests, and controlling system-level settings related to external information retrieval and account management.

2.4 Operating Environment

The system is expected to operate in the following environment:

- A mobile environment for general users
- A web-based environment for manager access
- Modern web browsers or mobile devices with internet access
- A server-connected environment for storing and delivering campus information
- Use of location services (GPS) for map-related functions

2.5 Design and Implement Constraints

The system is subject to the following constraints:

- The system must provide information through a map-based interface.
- The system should support marker and child marker structures for detailed location representation.
- General users should be able to use the main service without login.
- Only authorized managers should be allowed to log in and manage dynamic content through the web interface.
- Time-sensitive information such as dining, bus, or event detail is updated periodically and may not reflect real-time changes instantly.

- Some information may depend on external sources or manual updates by managers.
- The system shall support two languages: Korean and English, for the user-facing interface and marker information where applicable.

2.6 User Documentation

This system is intended for novice users and is designed to be immediately usable without prior training. Accordingly, it emphasizes an intuitive user interface so that users can understand and use its functions **without the need for separate documentation**.

2.7 Assumptions and Dependencies

The system is developed under the following assumptions and dependencies:

- Users have access to a device with **internet connectivity**.
- General users will access the service without authentication.
- Managers will use a web-based interface to log in and manage content.
- Dynamic information depends on periodic updates rather than continuous real-time synchronization.
- Some information may rely on official campus resources, external links, or manager input.

3. External Interface Requirements

3.1 User Interfaces

The system provides two user interfaces:

- **User Application** — a native mobile application for general Users (freshmen, exchange students, visitors) who access the service without login.
- **Manager Web Interface** — a web-based management console for authorized Managers.

To maximize development efficiency and reuse UI components across both interfaces, we recommend the user application to be developed as a web application first. The native mobile application may then be delivered as a **WebView wrapper** around the same web frontend.

Detailed UI differences between Users and Managers are described in the below sections.

3.1.1 User Application

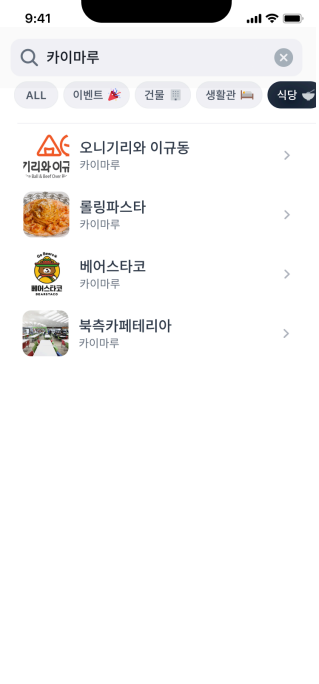
- 1) Main Page — Map, Marker, and Detail Sheet
 - a) **1-1:** On initial load, the map is panned to the user's GPS location. Markers are displayed on the map.
 - b) **1-2:** When a marker is tapped, the map focuses on the selected marker and the marker is highlighted. A bottom sheet appears showing the place's photo, title, title of parent marker, and Markdown-rendered information. If the selected marker is a parent marker, its child markers are listed below.
 - c) **1-3:** The bottom sheet can be expanded or collapsed by vertical touch-drag.
 - d) **1-4:** When a child marker is selected, the map focuses on that marker and its **static** information is displayed.
 - e) **1-5:** When a child marker with **dynamic** information is selected, such as a dining marker, the crawled information of the linked external web page and the result is rendered in the bottom sheet.
 - f) **1-6:** If the selected marker is categorized as a bus marker, the system fetches the nearest arrival time for each bus from the server and displays the remaining time until arrival.
 - g) **1-7:** If the selected marker is categorized as an event, the system displays the remaining time until the event ends.

1-1, Initial map centered on user's GPS with location markers	1-2, Marker tapped — bottom sheet shows photo, title, info, child marker list	1-3, Bottom sheet expanded via touch-drag	1-4, Child marker (static) — rendered info displayed

1-5, Dining Marker — server-crawled info displayed	1-6, Bus marker — next bus arrival times shown	1-7, Event marker — remaining time until event ends shown

2) Filter and Search of Markers

- a) **2-1:** When the map loads, markers are shown on the map. The top UI contains a search field and filter chips.
- b) **2-2:** When the dining filter chip is selected, only markers in the `dining` category are shown on the map.
- c) **2-3:** When a search query (e.g., `카이마루`) is entered, markers are filtered by the marker's title and parent marker's title.
- d) **2-4:** When a marker is selected from the search result list, the map focuses on that marker with highlights and the bottom sheet opens.

			
2-1, Initial map with search field and filter chips in top UI	2-2, Dining filter chip applied — only restaurant markers visible	2-3, Search query `카이마루` entered — results filtered by title/parent marker's title	2-4, Result selected from list — map focused, bottom sheet opens
			
This is an example of a filter component, and developers can customize to handle various edge cases.			

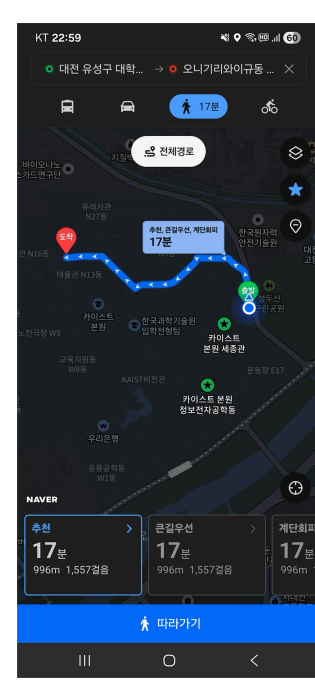
3) Marker Visibility Strategy

- a) **3-1:** At **zoomed-out level**, only parent markers are shown; child markers are hidden. This pruning reduces the number of visible markers, improving user experience and rendering performance.
- b) **3-2:** At **zoomed-in level**, all markers — including both parent and child markers — are shown.



4) Deep Link for Navigation Map

- a) **4-1:** When a marker is selected, the bottom sheet appears with a ‘길 찾기’ button (Navigation Map Button) at the bottom.
- b) **4-2:** When the button is clicked, the system opens an external map application's URL scheme for navigation routing, passing the user's GPS as the start point and the marker's coordinates as the destination.

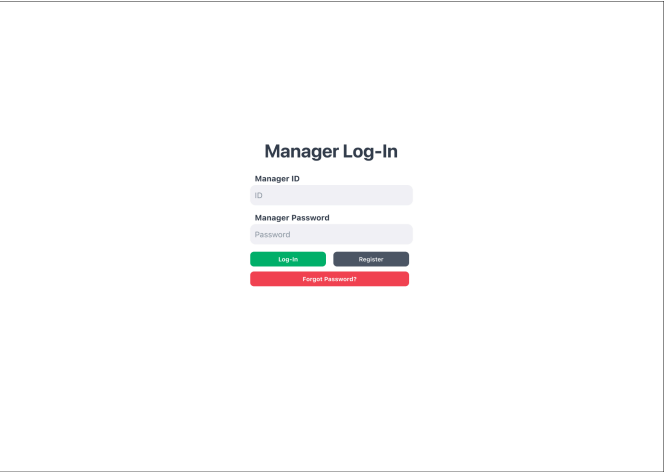
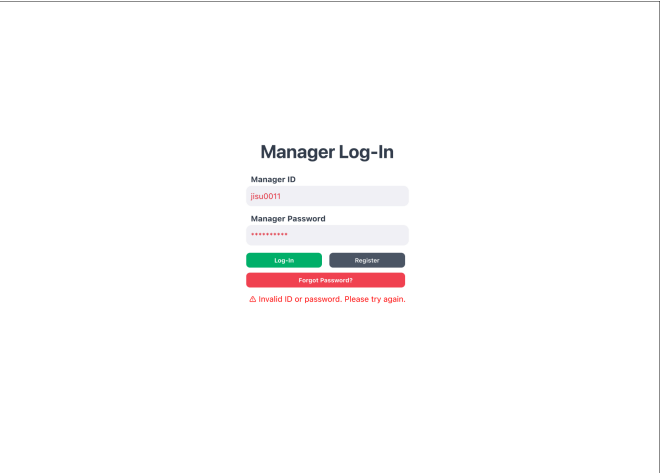
	
4-1, Bottom sheet with Navigation Map button at the bottom	4-2, External navigation app launched via URL scheme

3.1.2 Manager Marker Management System

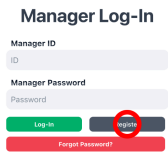
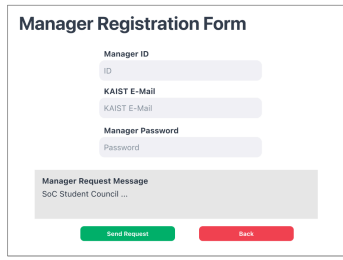
1) Manager Authentication / Registration

- a) **1-1:** When the manager login page is opened, the system displays input fields for manager ID and password, along with **Log-In**, **Register**, and **Forgot Password?** actions.
- b) **1-2:** When valid credentials are entered and the Log-In button is clicked, the system authenticates the manager and redirects the user to the manager marker management page.

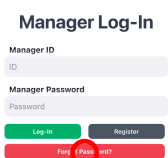
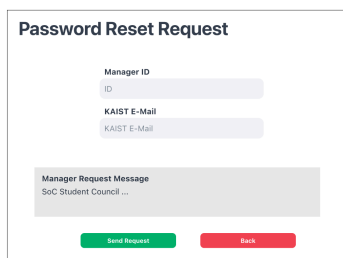
When invalid credentials are entered, the system remains on the login page and displays an error message indicating that the ID or password is incorrect.

	
1-1, Default manager login page with ID/password fields	1-2, Login Validation — valid login redirected to manager marker management page; invalid login triggering error message display on the page

- c) **1-3:** When the Register button is clicked, the system opens the manager registration request form.
- d) **1-4:** In the registration request form, the manager enters a desired ID, KAIST email address, password, and an optional request message, and submits the request to the administrator.

	
1-3, Register button clicked — registration request form opened	1-4, Registration request form submitted to administrator

- e) **1-5:** When the Forgot Password? button is clicked, the system opens the password reset request form.
- f) **1-6:** In the password reset request form, the manager enters identification information and submits the request to the administrator for manual review.

	
1-5, Forgot Password? clicked — password reset request form opened	1-6, Password reset request submitted to administrator

2) Filter of Marker List and Map

- a) **2-1:** When the manager marker management page is opened, the system displays the full marker list and corresponding markers on the map. The table header provides dropdown-based filtering controls for marker attributes such as category, activity status, and ownership.

- b) **2-2:** When the manager clicks the filter control in a table header, the system highlights the selected column and opens a filter interaction for that attribute.
- c) **2-3:** When the category filter is opened, the system displays a dropdown panel containing available category options. The manager can inspect the list of categories and select one or more values to filter the marker list.
- d) **2-4:** When one or more category values are selected, the system filters the marker list so that only markers matching the selected categories remain visible. At the same time, the map is updated to show only the corresponding markers.
- e) **2-5:** After the filter is applied, the table and the map remain synchronized. Only the filtered marker subset is shown, while non-matching markers are hidden from both the marker list and the map.

Marker Management

Select a marker to view or edit it.
New marker can be created with 'Create New Marker' button.

Create New Marker

Marker Title	Category	Active?	Mine?
Basketball Court (Sports Complex)	체육시설	Active	Mine
Bank (W8)	은행/ATM	Active	Others
Colloquium (E4)	이벤트	Inactive	Mine
...	...	...	...

2-1, Initial marker management page with full marker list and synchronized map

Marker Management

Select a marker to view or edit it.

New marker can be created with 'Create New Marker' button.

Search Markers

Create New Marker

Marker Title	Category	Active?	Mine?
Basketball Court (Sports Complex)	체육시설	Active	Mine
Bank (W8)	은행/ATM	Active	Others
Colloquium (E4)	이벤트	Inactive	Mine
...	...	...	...



2-2, Table-header filter control selected

Marker Management

Select a marker to view or edit it.

New marker can be created with 'Create New Marker' button.

Search Markers

Create New Marker

Marker Title	Category	Active?	Mine?
Basketball Court (Sports Complex)	Category	Active	Mine
Bank (W8)	☒ 체육시설	Active	Others
Colloquium (E4)	☐ 은행/ATM	Inactive	Mine
...	☐ 이벤트	...	...



2-3, Category dropdown opened with selectable category options

Marker Management

Select a marker to view or edit it.
New marker can be created with 'Create New Marker' button.

Search Markers

Create New Marker

Marker Title	Category	Active?	Mine?
Basketball Court (Sports Complex)	Category	Active	Mine
...	☒ 체육시설 ☐ 은행/ATM ☐ 이별트	...	...

2-4, Category filter selected — list and map updated with matching markers only

Marker Management

Select a marker to view or edit it.
New marker can be created with 'Create New Marker' button.

Search Markers

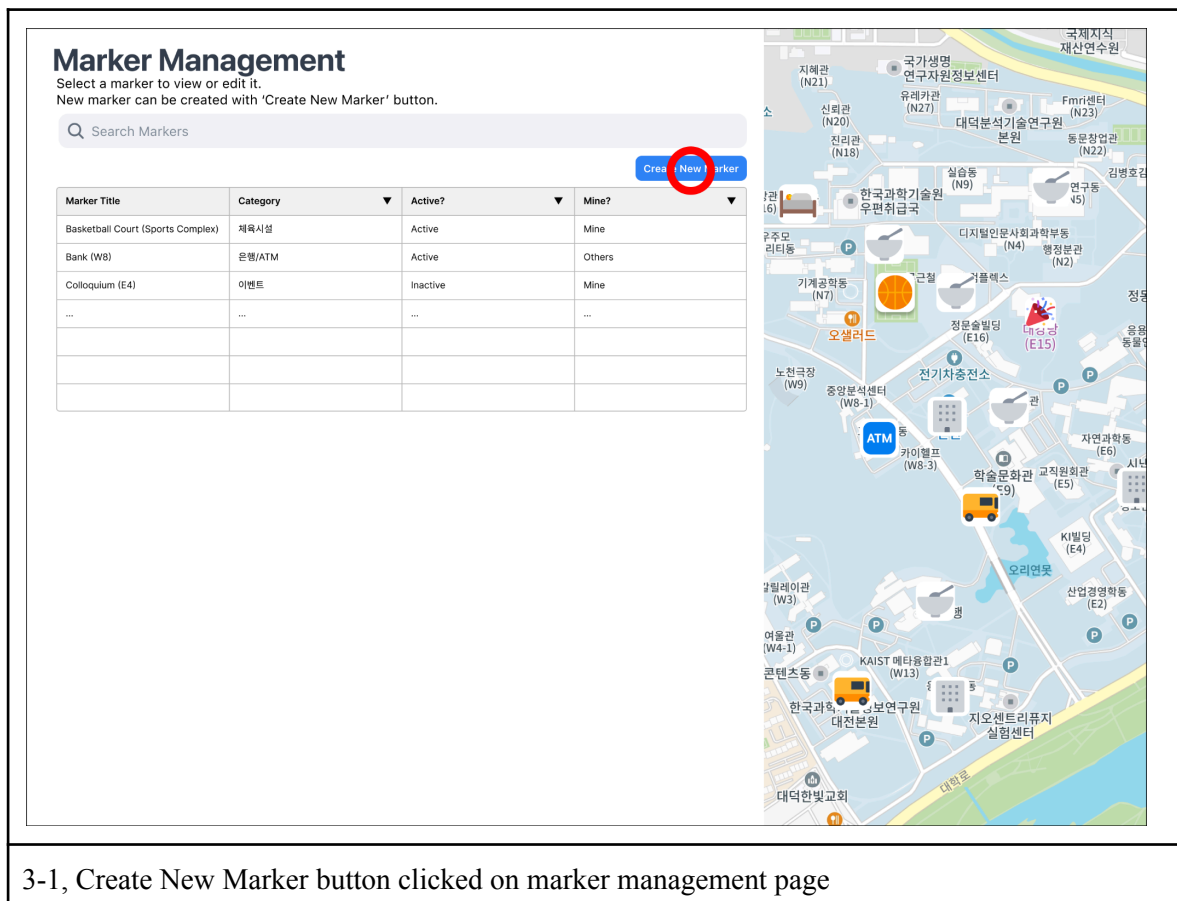
Create New Marker

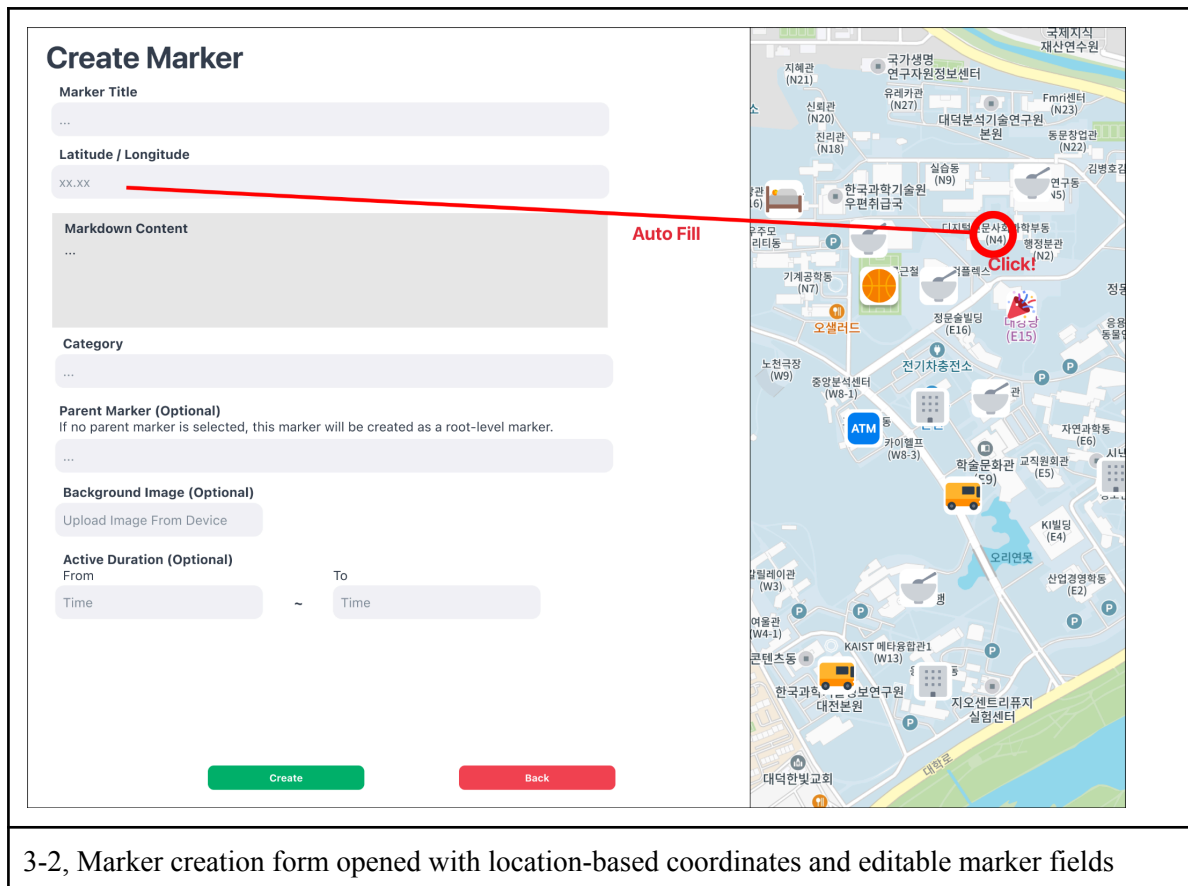
Marker Title	Category	Active?	Mine?
Basketball Court (Sports Complex)	체육시설	Active	Mine
...	...	...	...

2-5, Filtered result state — synchronized reduced marker set shown in table and map

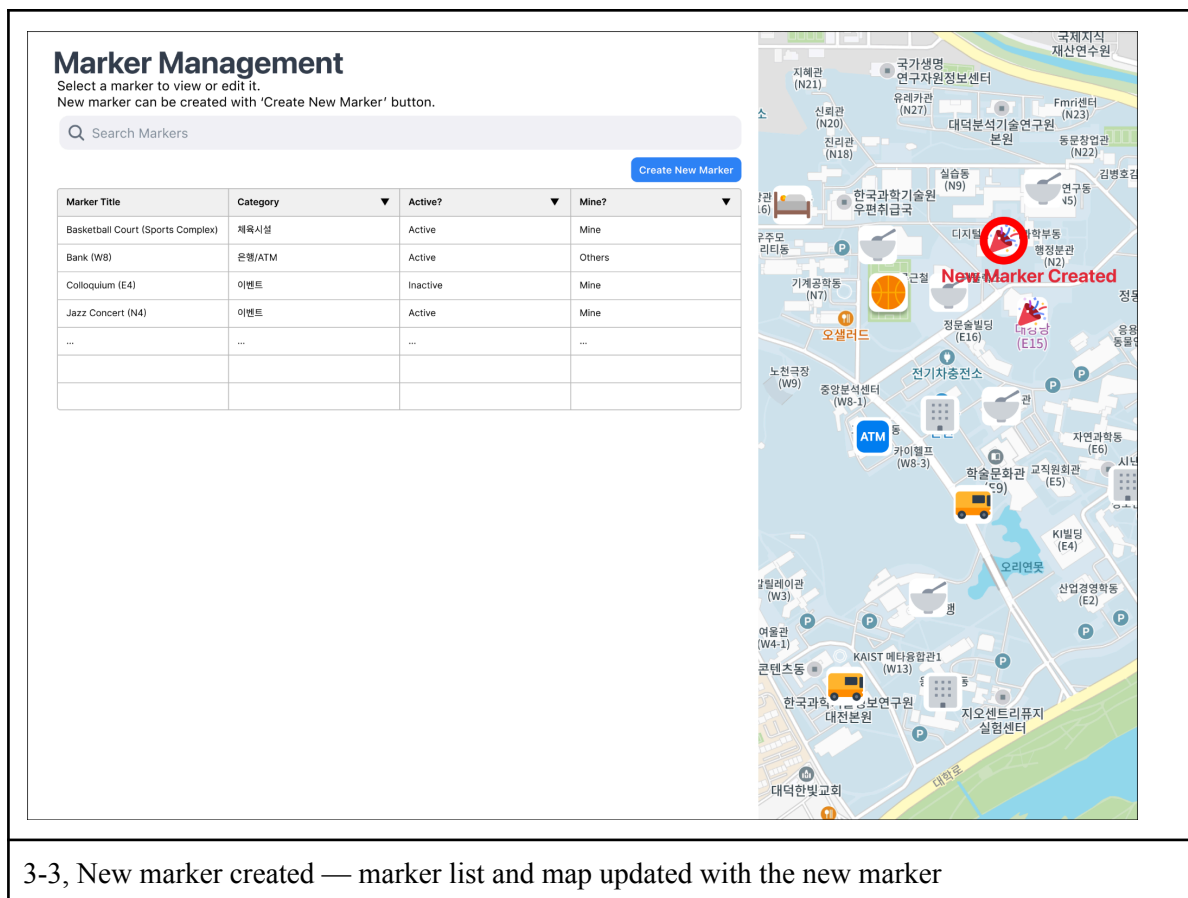
3) Marker Creation

- a) **3-1:** When the manager clicks the **Create New Marker** button on the marker management page, the system opens the marker creation form while keeping the map visible for location reference.
- b) **3-2:** In the marker creation form, the manager enters marker information including marker title, latitude/longitude, Markdown content, category, optional parent marker, optional background image, and optional active duration (only available for 'Event' category). The selected map location is reflected in the coordinate field.
- The markers can be turned into **Inactive Markers** when their active durations are outdated. These **Inactive Markers** should not be displayed on the map, but should remain on the list.
- c) **3-3:** When the manager completes the form and clicks **Create**, the system creates the new marker, updates the marker list, and renders the new marker on the map.





3-2, Marker creation form opened with location-based coordinates and editable marker fields



3-3, New marker created — marker list and map updated with the new marker

4) View and Edit of Self-Created Marker

- a) **4-1:** When the manager selects a marker row marked as **Mine** from the marker list, the system opens the marker detail view for that marker. The selected marker is also focused and highlighted on the map.
- b) **4-2:** In the marker detail view, the system displays the marker's stored information, including marker title, coordinates, Markdown content, category, optional parent marker, optional background image, and optional active duration(exclusive for event). Since the selected marker belongs to the current manager, the interface provides editing-related actions entry.
- c) **4-3:** When the manager clicks the edit action in the marker detail view, the system opens the marker edit interface with the existing marker values pre-filled in editable form fields. The manager can then modify the marker information and submit the changes.

Marker Management

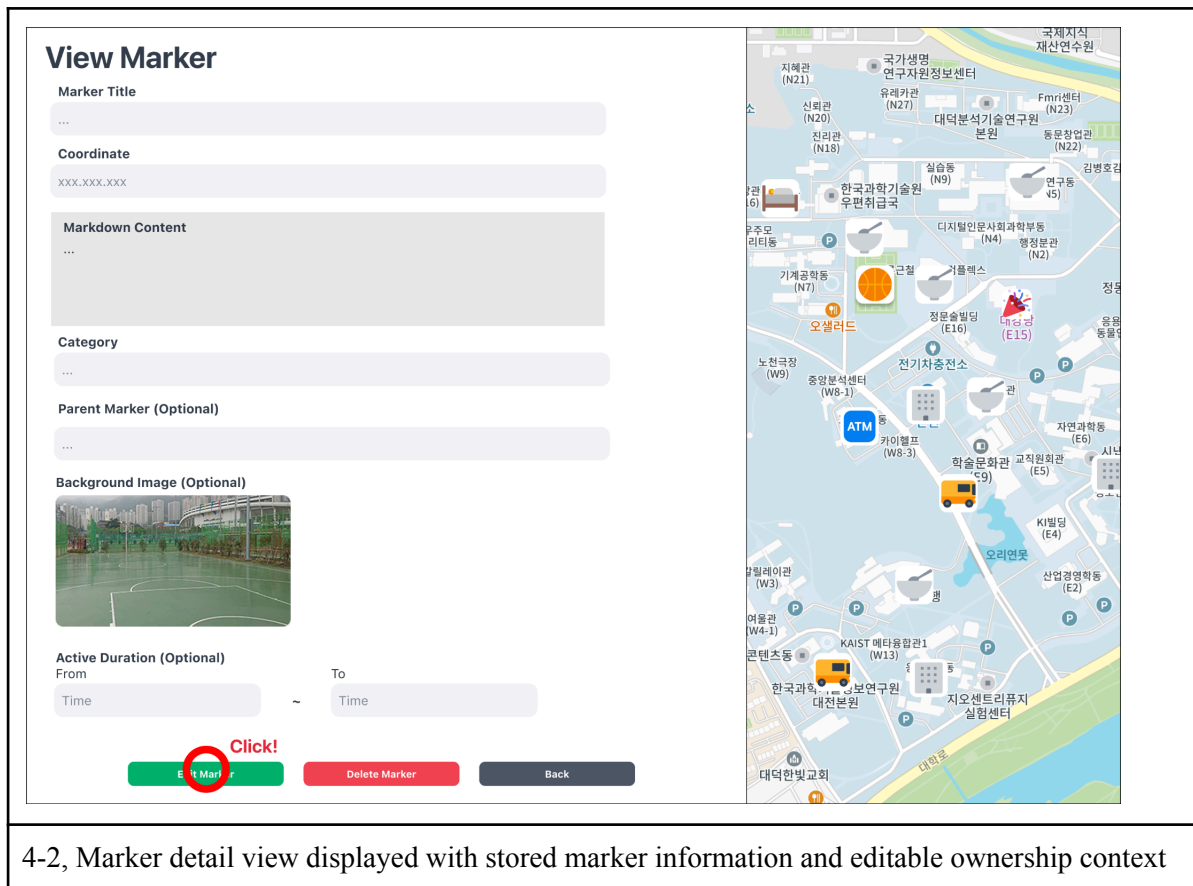
Select a marker to view or edit it.
New marker can be created with 'Create New Marker' button.

Create New Marker

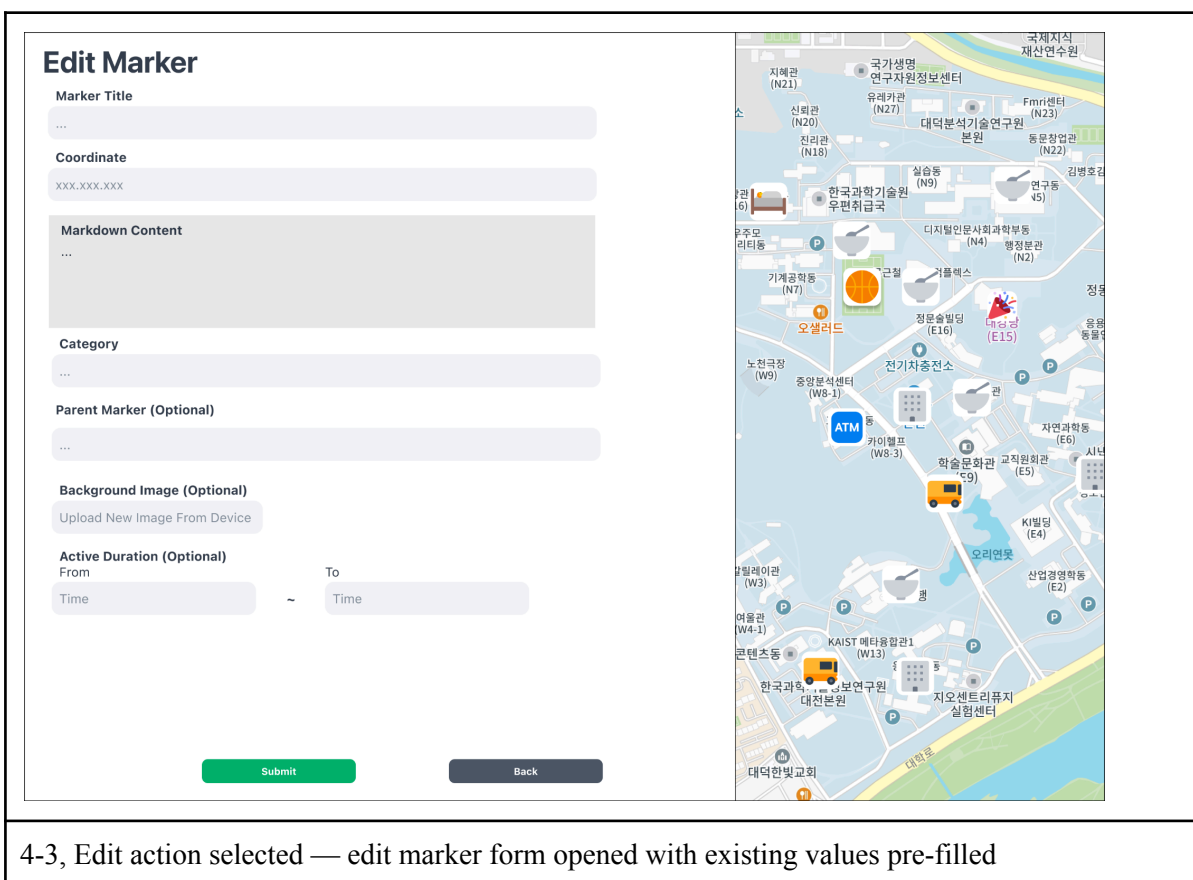
Marker Title	Category	Active?	Mine?
Basketball Court (Sports Complex)	체육시설	Active	Mine
Bank (W8)	은행/ATM	Active	Others
Colloquium (E4)	이벤트	Inactive	Mine
...	...	...	...

Click!

4-1, Self-created marker selected from marker list — detail view opened



4-2, Marker detail view displayed with stored marker information and editable ownership context



4-3, Edit action selected — edit marker form opened with existing values pre-filled

5) View of Other Manager's Marker

- a) **5-1:** When the manager selects a marker row marked as Others from the marker list, the system opens the marker detail view for that marker and focuses on and highlights the selected marker on the map.
- b) **5-2:** In the marker detail view, the system displays the marker's stored information, including marker title, coordinates, Markdown content, category, optional parent marker, optional background image, and optional active duration.

Because the selected marker was created by another manager, the system provides a read-only view only. Edit and delete actions are not available, and the manager can only return to the marker management page.

Marker Management

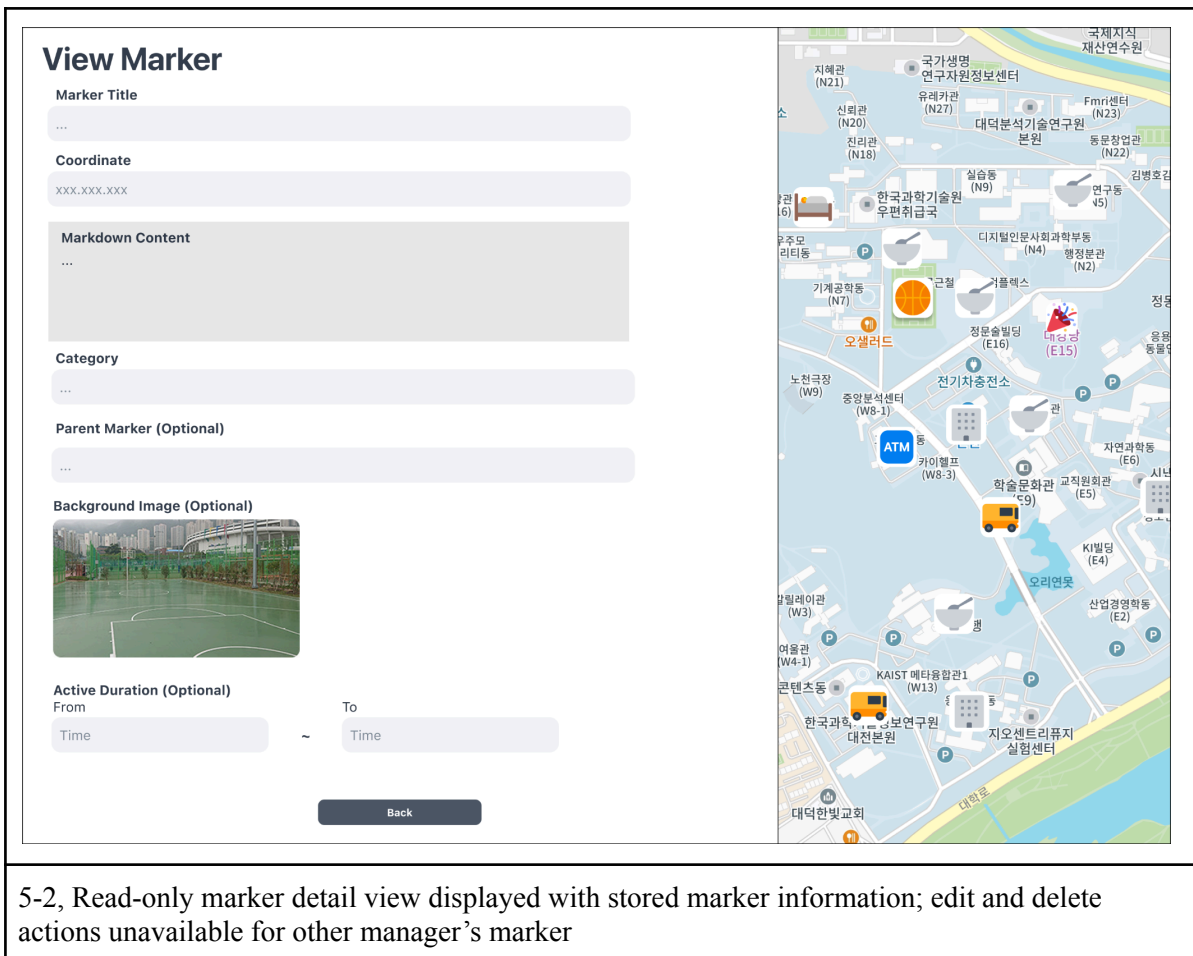
Select a marker to view or edit it.
New marker can be created with 'Create New Marker' button.

Search Markers

Create New Marker

Marker Title	Category	Active?	Mine?
Basketball Court (Sports Complex)	체육시설	Active	Mine
Bank (W8)	은행/ATM	Active	Others
Colloquium (E4)	이벤트	Inactive	Mine
...	...	...	...

5-1, Marker created by another manager selected from marker list



3.2 Hardware Interfaces

To ensure maximum accessibility for our users, the User Application is distributed as a native mobile application and is designed to support Android devices running version 9 or higher, with an SDK version of 28 or higher. This decision is based on the device distribution of our intended users. Additionally, we have made sure that the application is compatible with iOS devices running version 15 or higher, as this version or above is used by 92% of our users. The Manager Web Interface has no specific hardware constraints beyond a device with internet connectivity and a modern web browser.

3.3 Software Interfaces

3.3.1 Backend Server

To facilitate data exchange between clients and the database, the server provides APIs. For maximum accessibility and scalability, the server should be hosted within an AWS environment or another cloud platform. Additionally, the server should implement access control mechanisms tailored to each user's role, aligned with the user classes defined in this document — User, Manager, and

Administrator. By implementing these features, the server can provide secure and efficient data exchange while ensuring that each user has access only to the features and data they are authorized to use.

3.3.2 Database

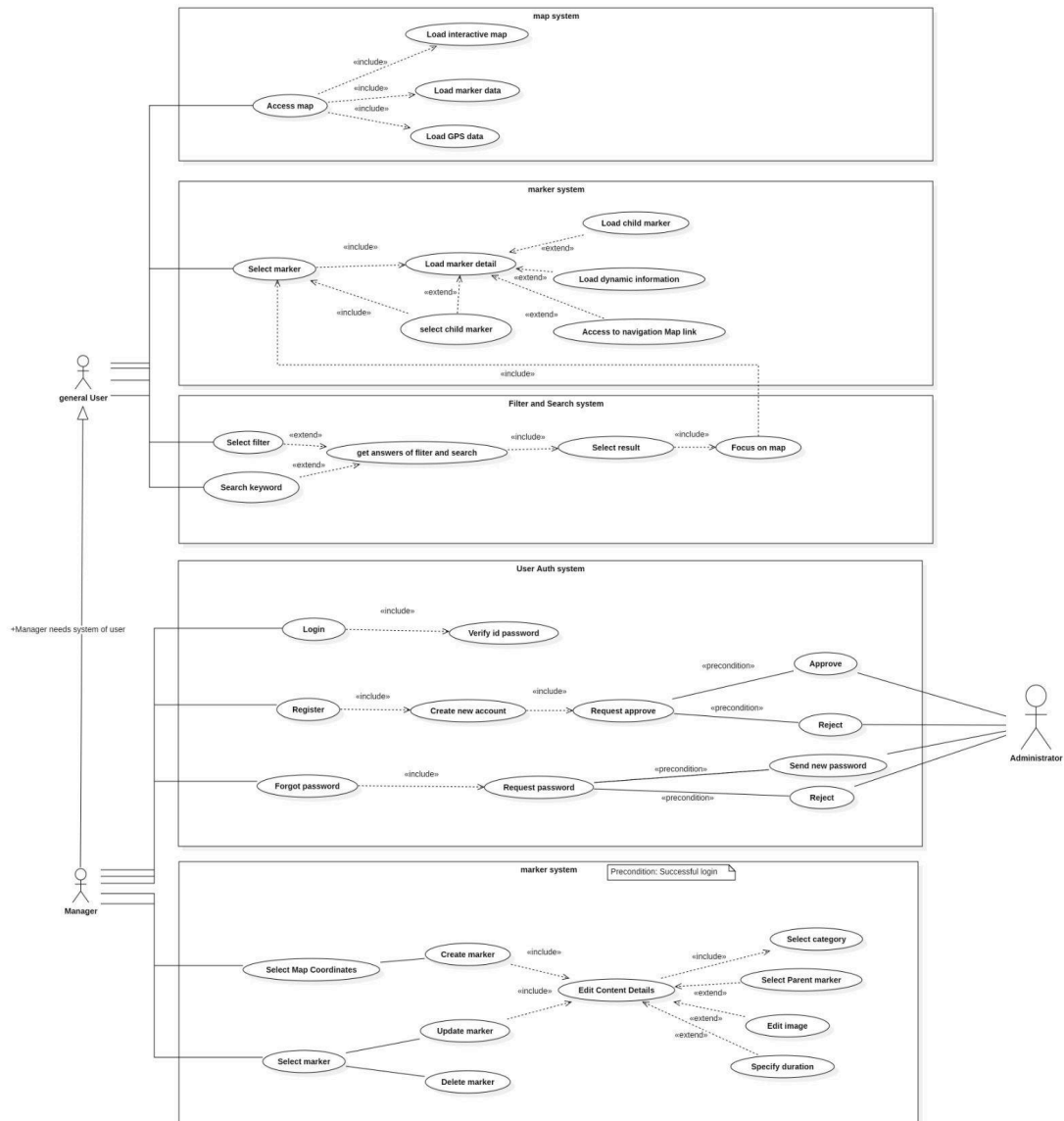
The database stores marker data and manager accounts in an organized manner, with a focus on maintaining referential integrity between related records such as parent-child marker relationships and manager-to-self-created-marker ownership. Each marker record contains title, Markdown-formatted content, a category, an optional parent marker reference, an optional background image. Each manager record contains credentials and approval status for authorized managers. The database shall also protect private information securely against external attacks or malicious exploitation.

3.4 Communication Interfaces

To provide feedback or make requests to the service manager, users can contact the designated email address posted on the service. For internal communication, the project consists of three main parts: the frontend, backend, and database. To transfer data between these components, the recommended format is JSON (either in array or object form) using REST APIs over the HTTPS protocol.

4. System Features

4.0 Whole use case model



4.1 Map View

4.1.1 Description

The Map View is the primary interface of the service, providing an interactive map based on a Map API (Kakao Map is recommended). Upon entry, the system automatically loads markers data and the GPS information.

The interface and functionality differ between **General Users** and **Managers**:

- **General Users:** Inactive markers are hidden. To optimize performance and readability, child markers are not rendered at zoomed out level.
- **Managers:** Managers have two views of markers. The map view is literally the same with general users, and the list view shows all markers in this system.

Detailed UI differences for each role can be found in the **User Interfaces** section.

4.1.2 Stimulus/Response Sequences

1. Access Map Screen(general users)
 - a. Load the interactive map using the Map API.
 - b. Load marker data from the server (inactive markers excluded).
 - c. Load GPS location data.
 - d. child markers are not rendered at zoomed-out level.
2. Access Map Screen (Managers)
 - a. Load the interactive map using the Map API.
 - b. Load marker data from the server.
 - c. Load GPS location data.
 - d. Managers have two views of markers. The map view is literally the same with general users, and the list view shows all markers in this system.

4.1.3 Functional requirements

REQ-4.1.a The system shall load and render an interactive map using the Map API upon entering the Map screen.

REQ-4.1.b The system shall load marker data from the server on map entry.

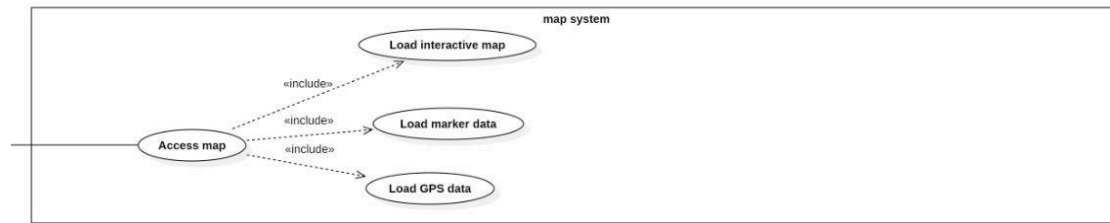
REQ-4.1.c The system shall retrieve and display the current GPS location on the map.

REQ-4.1.d General users and Managers shall be able to change the map zoom level by pinching or using zoom controls. The map shall respond to zoom level changes in real time.

REQ-4.1.e (General User) The system shall not display child markers at zoomed-out level. When the zoom level changes, the set of visible markers shall be updated accordingly.

REQ-4.1.f (Manager) For Managers, they have two views for markers, the map view shows the same with General users, and the list view shows all of the markers in the system.

REQ-4.1.g The system shall provide a distinct UI for General users and Managers, tailored to their respective needs.



4.2 Select Marker

4.2.1 Description

When general users or managers select a marker, the system retrieves and displays detailed information for that marker. This includes the marker's stored detail data as well as any supplementary dynamic information saved on the server (e.g., dining, bus).

Examples of dynamic information include the following:

- **Bus arrivals:** When getting data from the server, the marker should render the arrival time of the next bus for individuals and external link of bus arrival time table.
- **Event:** If an active duration is set for the marker, the system shall display the remaining time until the active duration expires.
- **Dining:** When getting data from the server, the marker should render the meal of today.

The interface and functionality differ between **General Users** and **Managers**:

- **General Users:** The combined information is rendered as Markdown on the frontend, It has the button for Directions, it should be connected to an external link with the GPS location for start, and marker location for arrival.
- **Managers:** Managers can access the marker details by selecting a marker from the list view. Managers can see the details of Marker with the view marker page by field. If the marker is a self-created marker, the manager can access the Edit page with the Edit button.

Detailed UI differences for each role can be found in the **User Interfaces** section.

4.2.2 Stimulus/Response Sequences

1. View Marker Detail (General Users)
 - a. The general user taps a marker on the map.
 - b. The system retrieves the marker's detail data and any associated dynamic information from the server.
 - c. The combined information is rendered as Markdown in the detail panel.

- d. Child markers linked to the selected marker are loaded.
 - e. If the general user taps a child marker, the detail view navigates to that child marker's detail.
 - f. If the general user taps a navigation button, the general user should be redirected to a Deep link. (we recommend Kakao Map).
2. View Marker Detail (Managers)
 - a. The manager taps a marker on the list.
 - b. The map view is focused on the selected marker.
 - c. The system retrieves the marker's detail data and any associated dynamic information from the server.
 - d. The information is displayed as a table in the View marker page.
 - e. Child markers linked to the selected marker are loaded.
 - f. If the Manager taps a child marker, the view marker page navigates to that child marker's detail.
 - g. If the marker is a self-created marker, the manager can access edit mode with the edit button.

4.2.3 Functional requirements

REQ-4.2.a The system shall retrieve the selected marker's detailed information and any stored dynamic information (e.g., menu, period, bus schedule) from the server upon selection.

REQ-4.2.b The system shall render the retrieved marker detail and dynamic information as Markdown (General Users) or by field (Managers) in the detail panel on the frontend.

REQ-4.2.c The system shall load and display child markers associated with the selected marker within the detail view.

REQ-4.2.d The general users and Managers shall be able to select a child marker from the detail view, which shall navigate to that child marker's detail view.

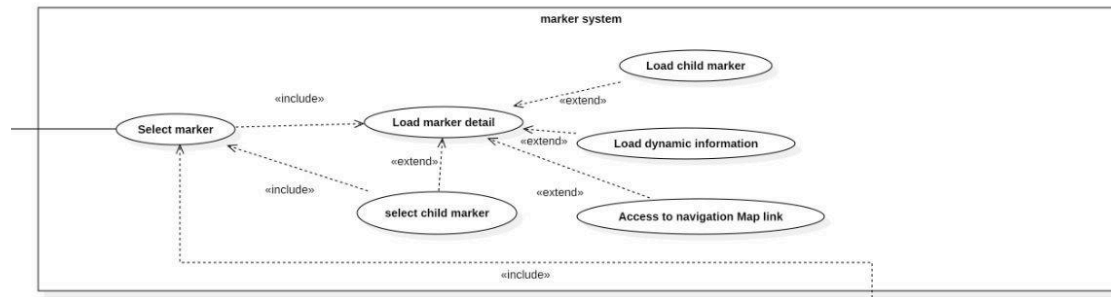
REQ-4.2.e (General Users) The system shall provide a navigation map button that connects to an external Map API (e.g. Kakao Map) via a URL scheme, passing the general user's GPS and the marker's coordinates.

REQ-4.2.f (Managers) The system shall provide an edit button on the View marker page only when viewing a self-created marker.

REQ-4.2.g (Dynamic information) The system shall display time-sensitive information based on marker category:

- **Bus:** Next bus arrival times.
- **Events:** Remaining time until the event ends.
- **Dinings:** The menu for the current day.

REQ-4.2.h The system shall provide a distinct UI for general users and Managers, tailored to their respective needs.



4.3 Filter And Search

4.3.1 Description

The Filter feature allows general users and managers to narrow down visible markers on the map or list by selecting one or more filter components. And the search feature allows general users and managers to find markers by title and optionally filter results. Search results are presented in a list. Selecting a result focuses the map on the corresponding marker and then transitions to the Select Marker flow.

The interface and functionality differ between **General Users** and **Managers**:

- **General Users:** General Users can filter markers only by category, such as dining and building. Filtered markers are displayed regardless of the current zoom level, overriding the zoom-based child marker rule.
- **Managers:** Managers can filter markers by category, ownership, and activation status. Manager filter and search apply to the list view and map view both.

Detailed UI differences for each role can be found in the **User Interfaces** section.

4.3.2 Stimulus/Response Sequences

1. Apply Search and Filter (General Users)
 - a. The general user selects one or more categories.
 - b. The system filters and displays only markers matching the selected categories.
 - c. Filtered markers should show for any zoom level.
 - d. The general user enters a search keyword.
 - e. The system searches for markers whose title or parent's title matches the keyword and satisfies the selected category filter.
 - f. Matching results are displayed in a dropdown list below the search bar.

- g. The general user selects a result from the dropdown.
 - h. The map focuses on the selected marker's location. (zoom in)
 - i. The system proceeds with the Select Marker flow. (see 4.2)
- 2. Apply Search and Filter (Managers)
 - a. The manager selects the filters on the list. (category, activation, create by own)
 - b. The system filters and displays marker info matching the selected categories on the list.
 - c. The manager enters a search keyword.
 - d. The system searches for markers whose category match the keyword and satisfy the selected filter.
 - e. Matching results are displayed in a list view.
 - f. The manager selects a result from the list.
 - g. The map focuses on the selected marker's location.
 - h. The system proceeds with the Select Marker flow. (see 4.2)

4.3.3 Functional requirements

REQ-4.3.a The system shall search markers by title and parent marker title.

REQ-4.3.b The system shall display only markers matching both the search keyword and the selected filter.

REQ-4.3.c When no filter and search is active, all markers shall be displayed according to the default rules.

REQ-4.3.d Upon selecting a result, the map shall focus on the selected marker's location.

REQ-4.3.e After focusing on the selected marker, the system shall proceed with the Select Marker flow as described in section 4.2.

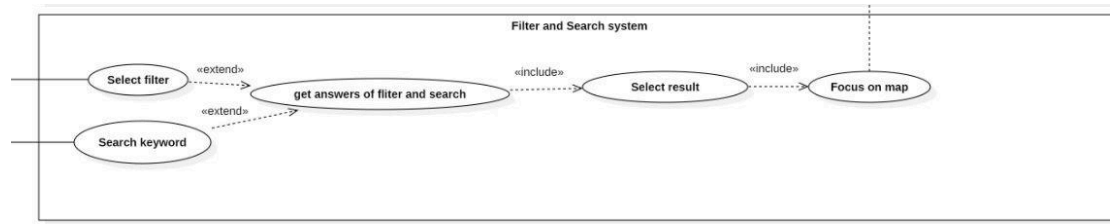
REQ-4.3.f If no markers match the search keyword and filter, the system shall display an appropriate message about indicating no results were found.

REQ-4.3.g (General Users) The system shall exclude inactive markers from the general user view.

REQ-4.3.h (General Users) When a category filter or search is active, the system shall display matching markers at all zoom levels, overriding the default child-marker visibility rule.

REQ-4.3.i (Manager) Manager filter and search shall apply to the list view and map view both.

REQ-4.3.j The system shall provide a distinct UI for general users and Managers, tailored to their respective needs.



4.4 Authentication System

4.4.1 Description

The Authentication System is exclusively for Managers. It consists of three flows: Login, Forgot Password, and Register. Login allows Managers to get access to marker management services. Forgot Password enables a Manager to request a password reset, which is processed manually by an administrator. Register allows a new Manager to request account creation, subject to administrator approval. General users may only access the map in a read-only capacity without logging in.

[Important] administrator tasks (e.g., account approval, password reset) shall be processed through admin, either via a dedicated interface or direct data management.

4.4.2 Stimulus/Response Sequences

1. Login
 - a. The manager enters their unique ID and password into the login form.
 - b. The system authenticates the credentials and redirects to the Map page on success.
 - c. On failure, the system clears the input fields and remains on the Login page.
2. Forgot Password
 - a. The manager enters their identification information on the Forgot Password page.
 - b. The system sends a password reset request to the Administrator.
 - c. Administrator reviews the request and sends either a new password or a rejection notice to the Manager's registered email.
3. Register
 - a. Manager clicks the 'Register' button on the Login page and is redirected to the Register page.
 - b. The manager enters their desired ID, password, and KAIST email address.
 - c. Manager clicks 'Register' to submit the account creation request to the Administrator.
 - d. The administrator approves or rejects the request; the applicant is notified by email in either case.
 - e. Upon approval, the Manager account is activated and the Manager may log in.

4.4.3 Functional requirements

REQ-4.4.a The Login page shall require the Manager to enter a unique ID and password.

REQ-4.4.b The system shall authenticate the Manager's credentials and redirect to the Map page upon successful login.

REQ-4.4.c On failed login, the system shall clear the input fields and display an error message on the Login page.

REQ-4.4.d The Forgot Password flow shall accept the Manager's ID and forward a password reset request to the Administrator.

REQ-4.4.e The Administrator shall be able to send a new password or a rejection notice to the Manager's registered email.

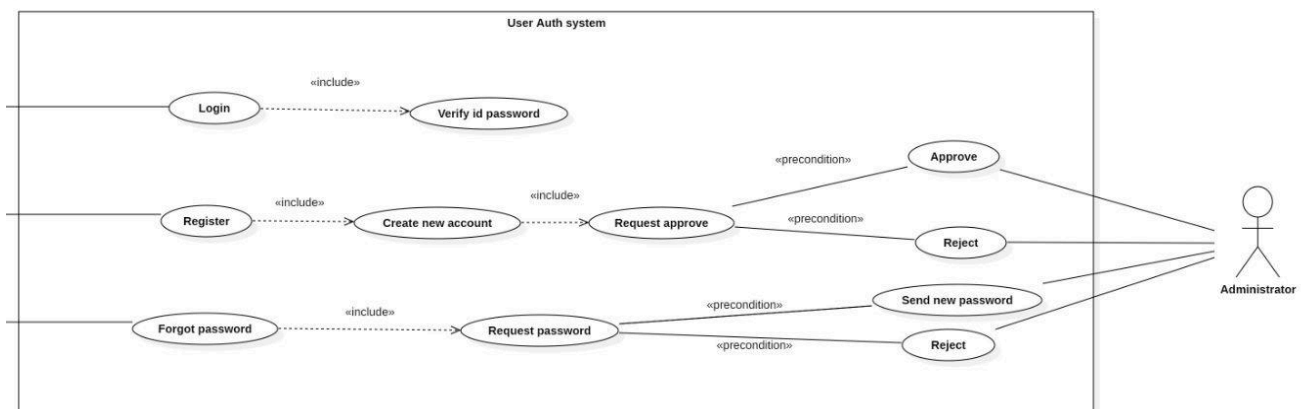
REQ-4.4.f The Register page shall require the general user to enter an ID, password, and KAIST email address.

REQ-4.4.g Upon registration submission, the system shall send an account creation request to the Administrator for approval.

REQ-4.4.h The system shall notify the applicant by email upon Administrator approval or rejection of the account request.

REQ-4.4.i A newly registered Manager account shall only be accessible after Administrator approval.

REQ-4.4.j (Option) Administrator tasks (e.g., account approval, password reset) shall be processed through administrator, either via a dedicated interface or direct data management.



4.5 Marker Management System

4.5.1 Description

The Marker Management System is available to Managers only and allows the creation, modification, and deletion of markers directly on the map. Each marker stores Markdown-formatted

content, a category, an optional parent marker, an optional background image, an optional active duration.

4.5.2 Stimulus/Response Sequences

1. Create a Marker
 - a. The manager clicks the 'add' button to create the marker.
 - b. Manager assigns appropriate position the latitude and longitude, which are auto-filled by clicking the map UI.
 - c. Manager enters the marker's Markdown content.
 - d. The manager selects a category.
 - e. Manager optionally selects a parent marker. (defaults to root if not selected)
 - f. Manager optionally adds, changes, or removes a background image.
 - g. The manager optionally sets an active duration.
 - h. The manager clicks the 'Submit' button to save.
2. Update a Marker
 - a. Manager selects a self-created marker on the list.
 - b. Manager clicks edit button.
 - c. Manager edits one or more fields.
 - d. The manager clicks 'Submit' to apply the changes.
3. Delete a Marker
 - a. Manager selects a self-created marker on the list.
 - b. Manager clicks edit button.
 - c. The manager clicks 'Delete' to remove the marker.

4.5.3 Functional requirements

REQ-4.5.a The system shall allow Managers to initiate marker creation by add button and select the latitude and longitude by clicking the map UI.

REQ-4.5.b Each marker shall store Markdown-formatted content, a category, an optional parent marker reference, an optional background image, an optional active duration.

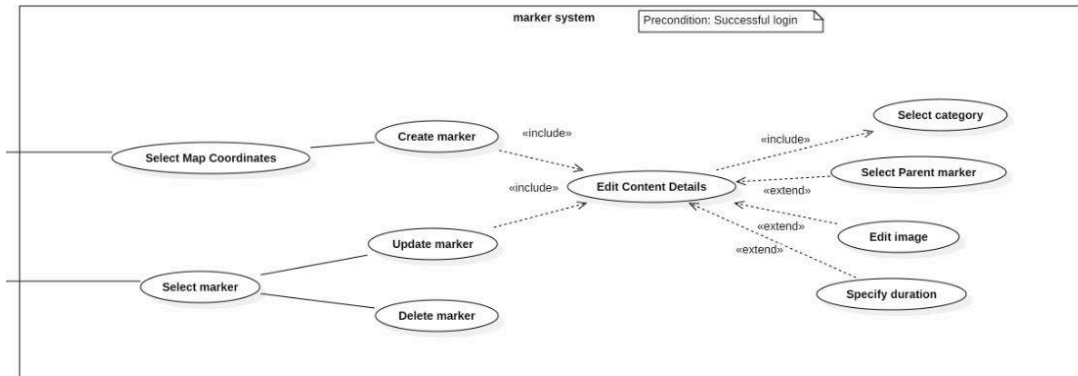
REQ-4.5.c If an active duration is set, the marker shall be visible to general users only during the specified active duration. Once the active duration has ended, the marker shall be hidden from the general user view.

REQ-4.5.d Clicking 'Submit' on the marker form shall create a new marker or update an existing one depending on context.

REQ-4.5.e Clicking 'Delete' on the marker form shall permanently remove the selected marker.

REQ-4.5.f Managers shall only have permission to update, delete for self-created markers.

REQ-4.5.g Crawling and retrieving external information cannot be added by the manager. If changes are required, they shall be handled by the administrator.



5. Other Nonfunctional Requirements

5.1 Performance Requirements

The system should provide a smooth and responsive experience for both general users and managers. Since the main service is based on an interactive campus map, the initial map screen should be displayed within a short time under normal network conditions, and visible marker data should be loaded without noticeable delay. Search results and filter updates should also be returned quickly so that general users can find locations and services efficiently while navigating the map. Marker detail information, including any associated dynamic data such as dining, bus, or event-related information, should be displayed promptly after a marker is selected. In addition, the system should support a reasonable number of simultaneous general users without critical failure of the core functions, especially the map view, search, and marker display features.

Manager-side updates such as marker creation, modification, or deletion should be reflected in the system within a short time after successful submission. This is especially important because the service is intended to provide practical information to general users who may already be moving around campus and need quick access to relevant details.

5.2 Safety Requirements

The system shall minimize confusion caused by inaccurate, incomplete, or outdated information. Since some information, such as dining, bus, and event-related content, is updated periodically rather than in real time, the system should indicate such limitations where appropriate. Markers with an active duration shall not be displayed to normal users outside the specified period. In addition, markers shown to users shall contain valid essential information, including location and descriptive content. The system shall also provide confirmation before managers perform critical actions such as marker deletion. These requirements are intended to ensure that users can rely on the service for practical campus guidance.

5.3 Security Requirements

Only approved managers shall be allowed to create, update, or delete dynamic marker information, and managers shall only be permitted to modify markers they created themselves. Manager registration shall require a valid KAIST email address, and newly requested accounts shall become active only after administrator approval. In addition, all manager inputs, including text, Markdown content, and URLs, shall be validated before storage. Administrative tasks such as account

approval, password reset handling, and external information retrieval settings shall remain under administrator's control.

5.4 Software Quality Attributes

Usability is the most important quality attribute of the system. Since many target users, such as freshmen, exchange students, and visitors, may be unfamiliar with the campus, the interface should be intuitive and easy to understand without prior training. The system shall also maintain reliability by presenting consistent and meaningful marker information. Maintainability is important so that marker data, categories, and dynamic information sources can be updated independently. In addition, the system should support extensibility for future categories or information sources and remain compatible with both mobile environments for general users and web environments for managers.

5.5 Business Rules

General users may access campus information without authentication, but they shall not modify any system data. Only approved managers may access the manager interface and manage dynamic marker information. A manager account shall become active only after administrator approval, and managers shall only be allowed to edit or delete markers they created. Markers with an active duration shall be visible to normal users only during the specified period. In addition, external information retrieval settings and other system-level configurations shall remain under administrator control. These rules define the separation between public information access and authorized content management.

6. Other Requirements

Appendix A. Glossary

Term	Description
General users	People who use the service without logging in. General users can browse the campus map, search for locations, apply filters, and view marker information.
Managers	Authorized users who log in to the management interface and manage marker information.
Administrator	The person responsible for administrative tasks such as approving manager accounts, handling password reset requests, and controlling system-level settings.
Marker	A location-based item displayed on the map. A marker represents a place, facility, stop, event, or other campus-related point of interest.
Marker title	The name or label assigned to a marker and used to identify it in the map view, search results, and management interface.
Parent marker	A marker that contains one or more related markers representing more detailed spaces or facilities within a larger location.
Child marker	A marker associated with a parent marker and used to represent a more specific location inside or under that parent marker.
Static information	Information stored in the system that does not change frequently, such as location descriptions, building information, or facility details.
Dynamic information	Information that changes over time and therefore may need periodic updates or retrieval. Dynamic information includes time-sensitive information such as event information, bus information, and dining information.
Event	A marker or marker-associated item created by a manager to represent temporary or time-sensitive campus information for a limited period. An event is one type of dynamic information in the system.
Inactive marker	A marker that is not currently shown to general users because it is no longer active. This includes event markers whose active duration or period has already passed. Depending on the system context, inactive markers may still be visible to managers in the management interface.
Active duration / period	The time range during which an event-related marker or temporary information is valid, visible, or considered active in the system.

Dining	Campus dining-related information, including dining locations, menus, meal plans, and related content. In this document, dining, restaurant, and cafeteria are treated under the concept of dining.
Bus	Bus-related information, including bus stop locations, arrival information, and related transportation content. In this document, campus shuttle (including OLEV) are treated under the concept of bus.
Category	A classification assigned to a marker so that markers can be organized, filtered, and searched by type. Examples include dining, bus, event, and other location categories.
Map view	The main map-based interface in which markers are displayed and users interact with campus information. Terms such as map screen, main page, and map page are treated as map view.
Detail view	The interface area in which the selected marker's information is displayed. Terms such as detail panel, detail sheet, bottom sheet, and view marker page are treated as detail view.
Search	A function that allows users to find markers by entering keywords (title and parent marker title) related to marker title or related information.
Filter	A function that allows users to narrow visible markers based on selected conditions such as category, status, or ownership.
Ownership	The relationship between a manager and a marker indicating whether that marker was created by the current manager or by another manager.
Self-created marker	A marker created by the currently logged-in manager and therefore editable by that manager.
Map API	The external map service used to render the interactive campus map and support map-based functions.
GPS location	The current geographic location of the user's device used for centering the map or generating directions.
Navigation	A function that opens an external navigation service and passes the user's GPS location as the start point and the selected marker location as the destination.
External information	Supplementary information associated with a marker that may come from linked external sources or server-side retrieval processes.
Markdown content	Formatted text stored for a marker and rendered in the interface to present descriptive information.
Zoomed-in level	A map state in which the user has zoomed in and more detailed spatial information is shown. In this state, more markers, including detailed markers, may be displayed.

Zoomed-out level	A map state in which the user has zoomed out and a broader area is shown with less detail. In this state, some detailed markers may be hidden to reduce clutter and improve readability.
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